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Construal Level Theory and Persuasion on ChangeMyView

A Sentence-Level Classifier Cascade and Conditional Logistic Analysis in Economics Threads on ChangeMyView

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ABSTRACT

This thesis investigates whether psychological distance, as formalised by Construal Level Theory, affects the persuasive success of arguments in online discussion. The empirical setting is the ChangeMyView subreddit, in which users explicitly signal a change of view by awarding a Δ, providing a directly observable outcome for persuasion. A two-stage transformer cascade is trained on a hand-annotated subsample of 1,054 sentences to detect linguistic cues of temporal, spatial, social, and hypothetical distance, and the sentence-level predictions are aggregated to four comment-level abstraction indices. These indices are entered into a thread-stratified conditional logistic regression on the economics-filtered CMV corpus, with surface linguistic and structural features entered as adjustment covariates and a crossed-random-effects model as a robustness check. Classifier diagnostics revealed that the temporal and spatial heads collapsed to the dominant class on the held-out test set; the confirmatory analysis is therefore restricted to social and hypothetical distance, with temporal and spatial reported as exploratory. Net of surface features, concrete (near) construal is associated with greater persuasive success on both estimable dimensions, though the two results differ in robustness. The social effect is negative across every surface-adjusted specification, the direction theory predicts, but it reverses before adjustment and is therefore conditional on treating the surface features as confounders rather than mediators. The hypothetical effect is negative throughout and significant in the better-powered sample, the opposite of the standard Construal Level Theory prediction, although this dimension partly conflates psychological distance with epistemic stance. Both effects are small: the construal indices have almost no standalone discriminative power, and between-author and surface-feature variation dominates the abstraction signal. Within these limits, the findings suggest that in economics and policy argumentation on ChangeMyView, persuasion favours concrete over abstract framing; an exploratory test ties this to construal fit, with persuasive success tracking how closely a comment matches the original poster's construal. The current annotation scale constrains which distance dimensions can be recovered from argumentative text at all.

CONTENTS

PREFACE

This thesis was written in partial fulfilment of the requirements for the degree of Bachelor of Science in Information Science at the University of Groningen. The project sits at the intersection of psycholinguistics and computational text analysis, two areas I had the chance to combine for the first time in this work.

I would like to thank my supervisor Khalid Al-Khatib for the consistent feedback and patient guidance throughout the project, and for the willingness to engage with a topic that required pulling on both the theoretical and computational sides at once. I would also like to thank my fellow thesis students Sebastiaan Wegink and Patrick Jans, whose parallel work on the shared annotation effort and the joint classifier shootout made the empirical scope of this thesis possible.

Declaration on the use of AI tools. In the writing process of this thesis, AI tools have been used for assistance with academic writing, grammar, structural feedback, and code writing. All empirical analysis, data collection, interpretation of results, and academic arguments are the author's own work. AI-generated content has been critically reviewed and adapted by the author throughout. The use of AI tools complies with the guidelines of the University of Groningen.

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1 INTRODUCTION

Whether an argument persuades online depends on its framing as much as its content. People evaluate arguments through the way they mentally represent them: psychologically distant content tends to be processed abstractly, near content concretely (Trope and Liberman, 2010). Construal Level Theory provides the central framework for this distinction and argues that temporal, spatial, social, and hypothetical distance are all manifestations of psychological distance (Trope and Liberman, 2010).

This thesis examines whether psychological distance affects the persuasive success of arguments in online discussion. The empirical focus is the ChangeMyView subreddit, specifically threads classified as economics-related, a domain in which arguments tend to engage groups, institutions, and counterfactual policy scenarios. CMV is a large-scale discussion platform in which users explicitly signal that their view has changed by awarding a Delta (Δ) (Tan et al., 2016). This makes the platform particularly suitable for studying persuasion in natural language data, because success is observable rather than inferred indirectly (Tan et al., 2016). Prior research has shown that ChangeMyView can be used to study convincingness and argumentative interaction in online debates, but the role of psychological distance as a structured persuasive factor remains underexplored.

The relevance of this question is both theoretical and practical. Theoretically, it connects persuasion research with a psychological framework that has been widely used to explain how people interpret events across different levels of distance (Trope and Liberman, 2010). Practically, it may contribute to computational argumentation and natural language processing by identifying psychologically grounded features that help explain why some arguments succeed while others do not (Habernal and Gurevych, 2016).

This study addresses all four dimensions of psychological distance, temporal, spatial, social, and hypothetical, but, as detailed in Chapter 4, classifier performance and class sparsity restrict the primary inferential analysis to social and hypothetical; temporal and spatial are reported as exploratory. The thesis asks whether the level of construal expressed in an argument predicts persuasive success, and specifically whether near or far construal on the social and hypothetical dimensions is associated with a higher probability of receiving a Δ. In other words, it tests whether construal level is a linguistically detectable predictor of persuasion, net of surface features such as length and lexical complexity.

To answer this question, the thesis combines transformer-based text classification with statistical modelling, in three steps. First, a classifier is trained on a hand-annotated sample of CMV sentences to detect linguistic cues of psychological distance, with fine-tuned transformer encoders such as BERT (Devlin et al., 2019) and RoBERTa (Liu et al., 2019), and a prompted instruction-tuned LLM (LLaMA 3.2, Meta AI, 2024), among the candidates. Second, the sentence-level predictions are aggregated into one continuous abstraction index per psychological dimension, quantifying the level of construal in each comment. Third, these indices are entered into a conditional logistic regression, stratified by thread, that tests the main hypotheses; surface linguistic and structural features are added as adjustment covariates, and a mixed-effects model with crossed thread and author random effects serves as a robustness check. This design moves beyond surface text features and tests a theory-driven account of persuasion in online argumentation (Habernal and Gurevych, 2016; Trope and Liberman, 2010).

The contribution is methodological and theoretical. Methodologically, the thesis builds and validates a sentence-level cascade that recovers construal cues at scale and tests them against an observable persuasion outcome (Habernal and Gurevych, 2016). Theoretically, it shows that the classical Construal Level Theory prediction does not hold in this adversarial setting: net of surface features, concrete rather than abstract framing predicts persuasive success on the measurable dimensions (Trope and Liberman, 2010). An exploratory follow-up ties this to construal fit, the match between an argument's framing and the addressee's, and a structural check shows the distance dimensions are unevenly marked and do not reduce to a single construal level.

The remainder of this thesis is organised as follows. Chapter 2 reviews the theoretical foundation, covering Construal Level Theory (Trope and Liberman, 2010) and computational argumentation research on persuasion (Habernal and Gurevych, 2016). Chapter 3 describes the data: the ChangeMyView material drawn from the Webis-CMV-20 corpus (Kolyada et al., 2020), the hand-annotated subsample used to train the classifiers, and the construction of the matched analysis sample. Chapter 4 sets out the method, including the classifier shootout among fine-tuned transformer encoders (BERT and RoBERTa) and a prompted instruction-tuned LLM (LLaMA 3.2) (Devlin et al., 2019; Liu et al., 2019; Meta AI, 2024), the construction of the abstraction indices, and the specification of the conditional logistic regression. Chapter 5 reports the empirical findings: the selected classifier, its dimensional coverage, and the hypothesis tests, which point to a concrete-construal advantage rather than the abstract advantage the theory predicts. Chapter 6 discusses these findings for information science, sets out the limitations, and suggests directions for future work.

2 BACKGROUND

2.1 THEORETICAL BACKGROUND

This chapter reviews the main research areas relevant to this thesis: Construal Level Theory (CLT), the intersection of psychological distance and persuasion in language, and prior computational work on the ChangeMyView (CMV) corpus. The goal is to situate the present study within the existing literature and demonstrate how it extends previous models of argument quality (Habernal and Gurevych, 2016).

2.1.1 Construal Level Theory and Psychological Distance

Construal Level Theory (CLT) is the central psychological framework underlying this thesis. CLT proposes that people mentally represent events at different levels of abstraction depending on their perceived psychological distance (Trope and Liberman, 2010). Distant events tend to be construed more abstractly (high-level construals), whereas near events are construed more concretely (low-level construals) (Trope and Liberman, 2010).

Psychological distance is typically categorized into four dimensions: temporal, spatial, social, and hypothetical distance (Trope and Liberman, 2010). These dimensions are egocentric, anchored in the "here and now," and while they capture different ways an event may feel distant, they are functionally related in how they affect cognitive representation (Bar-Anan et al., 2007).

For CLT, abstraction is not just a property of style: it is part of how people represent meaning and judge information (Trope and Liberman, 2010). This matters for persuasion research, because readers respond both to the content of an argument and to how well its framing fits their perceived distance from the topic (cf. Greschner et al., 2025). CLT offers a cognitive account of mental representation, but applying it computationally requires a bridge between these internal states and their expression in language.

2.1.2 Construal Level and Persuasive Fit

Although CLT began as a theory of judgement and prediction, it has been applied directly to persuasion, where the dominant finding is one of *fit* rather than a uniform advantage for abstract or concrete framing. Kim et al. (2009) show across three experiments that abstract, "why"-oriented appeals are more persuasive when a decision is temporally distant, whereas concrete, "how"-oriented appeals are more persuasive when it is imminent, with the effect strongest among less-informed audiences. White et al. (2011) report the analogous pattern for message framing: loss-framed appeals persuade more under a concrete mind-set and gain-framed appeals under an abstract one. In both cases the persuasive gain comes from a match between the construal level of the message and the construal level at which the audience is operating, a match that raises processing fluency and produces a subjective "feels right" experience that transfers to the evaluation. The implication is that the effect of construal on persuasion is conditional rather than a main effect: concrete framing should hold the advantage precisely when the audience is reasoning about something psychologically near. ChangeMyView is such a setting: a direct, real-time debate between identified interlocutors, in which the fit account would predict a concrete advantage. The hypotheses pre-committed in this thesis (Section 4.2) nonetheless follow the classical, main-effect formulation of CLT; whether the fit account better describes the observed pattern is taken up in the Discussion (Section 6.1).

2.1.3 Psychological Distance in Language

Prior research has established that psychological distance is encoded within natural language. Language reflects abstraction through lexical choice, syntactic structure, and the level of specificity used to describe actions or events (Semin and Smith, 1999). The schema operationalises the four CLT distance dimensions directly as near/far levels; it is informed by the view that abstraction is linguistically encoded (Semin and Smith, 1999) but does not apply the Linguistic Category Model's interpersonal-verb coding, a difference that limits direct comparability with LCM-based work. Semin and Smith (1999) demonstrate that concrete language is associated with immediate, actionable content, while abstract language is linked to distant or hypothetical scenarios.

This makes language a promising medium for operationalising CLT computationally: distance cues can be detected at scale with modern Natural Language Processing (NLP) (Zymla et al., 2024). Detecting them moves analysis past surface text features to the cognitive framing beneath, and makes it possible to test construal-level effects in large, real-world argumentative settings such as ChangeMyView.

2.1.4 ChangeMyView as Empirical Setting

The empirical setting of this thesis is ChangeMyView (CMV), a Reddit community where users post opinions and invite others to challenge them (Tan et al., 2016). The platform is valuable for persuasion research because users explicitly award a Delta (Δ) when a comment changes their view, providing a clear, non-inferred outcome variable for persuasive success (Tan et al., 2016). Consequently, CMV has become a standard corpus for studying argument quality and interaction in online deliberation (Habernal and Gurevych, 2016).

Previous work on CMV has examined features such as argument length, lexical diversity, and interactional dynamics (Tan et al., 2016). These studies show that persuasive success is linked to identifiable patterns in language and discussion behavior (Habernal and Gurevych, 2016). However, much of this work has focused on these structural features rather than the psychologically motivated dimensions of distance proposed by CLT. Despite the wealth of data provided by the platform's unique reward system, existing research has yet to fully integrate the psychological principles of distance into models of persuasive success.

2.2 LITERATURE REVIEW

2.2.1 Structural Features of Persuasive Comments

Early studies on ChangeMyView (CMV) focused on the observable structural properties of comments, what a persuasive comment looks like. Althoff et al. (2014) set the foundation by analyzing thousands of threads, finding that Delta-awarded comments were systematically longer, used more bullet points, and included more external links. These features signal effort and credibility.

These are surface markers of effort, not explanations of why effort persuades. The interactional studies discussed next take up that question.

2.2.2 Interactional and Stylistic Dynamics

While structure provided the foundation, later research shifted toward the relational and stylistic dynamics of persuasion, how interlocutors actually interact. Tan et al. (2016) advanced this by introducing the concept of "interactional symmetry," showing that successful persuaders often mirror the original poster's vocabulary and emotional tone. This language style matching builds rapport, potentially making recipients more cognitively open to changing their stance.

Musi et al. (2018) and Jeong and Chiu (2024) extend this to interpersonal strategy, showing that concessions and polite discourse shape whether an argument succeeds. Such moves let a challenger respond directly to the opponent's framing rather than rely on word count or formatting alone (Messerli et al., 2025).

2.2.3 From Surface Features to Cognitive Mechanisms

Across these two strands, the structural and the interactional, the same limitation recurs: both describe what persuasive comments look like without explaining why those choices work cognitively. Althoff et al. (2014) and Tan et al. (2016) account for success through effort and alignment, but not through how an argument is mentally construed at different levels of abstraction (Trope and Liberman, 2010).

This thesis addresses that gap with Construal Level Theory. Where earlier work measured surface features (word count, hyperlinks) and behavioural alignment (style matching), CLT focuses on framing: whether a comment is pitched at a concrete, near level or an abstract, far one, and how that affects persuasion. Operationalising these dimensions lets the study test a cognitive account of persuasion rather than only describing surface patterns.

2.2.4 Research Gap and Contribution

This thesis extends previous work in two main ways. First, it applies CLT to a large-scale online persuasion corpus, allowing the theory to be tested in an ecologically valid environment rather than a controlled laboratory setting. Second, it combines transformer-based text classification with statistical modeling to operationalize psychological distance specifically within the domain of argumentative discourse (Messerli et al., 2025). In addition to the confirmatory test, the thesis adds an exploratory step that prior CMV work has not taken: it asks whether persuasive success depends on the *match* between a comment's construal and that of the original poster it answers, the alignment the fit account predicts (Kim et al., 2009; White et al., 2011).

In summary, while the literature suggests that psychological distance is a meaningful concept for understanding language, its interaction with argument success on CMV remains unexplored. The following chapter describes the data and methodology used to investigate this relationship.

3 DATA AND MATERIAL

This chapter describes the empirical material and how it was prepared. The study draws on three distinct samples, and keeping them separate matters for what each result can claim. The first is the full ChangeMyView corpus, from which the binary persuasion outcome is derived (Section 3.1). The second is a small hand-annotated subsample, used only to train and evaluate the construal classifiers (Section 3.2). The third is the matched analysis sample on which the regression is estimated, built from the full corpus after the classifier has labelled it (Section 3.3). Section 3.4 describes the text processing common to all three.

3.1 THE CMV CORPUS AND THE OUTCOME VARIABLE

The empirical material is drawn from the ChangeMyView (CMV) subreddit via the Webis-CMV-20 corpus (Kolyada et al., 2020), which provides submissions and comments together with the Δ-award metadata required for the binary outcome variable (Tan et al., 2016). The Δ-recipient is derived by parsing the username from DeltaBot's confirmation reply and matching it against the earliest ancestor comment in the same branch. Validation against the curated subset of Kolyada et al. (2020) yielded recall 0.93 on 5,990 overlapping threads; unmatched cases were dominated by comments with deleted authors, which are subsequently excluded under the deleted/removed filter. The outcome is defined per comment j in thread i as $y_{ij} = 1$ if the comment received a Δ and 0 otherwise. To eliminate cross-topic confounding, the corpus is restricted to threads classified as economics-related using an existing zero-shot topic classifier (Laurer, 2024; He et al., 2020), ensuring that distance cues are compared within a consistent argumentative domain. The classifier was validated on a hand-coded sample of 50 threads (seed 42), yielding 92% accuracy (precision 0.75, recall 1.00). The asymmetric error profile favours recall over precision, ensuring near-complete capture of true economics threads at the cost of admitting roughly a quarter as thematically adjacent (e.g., political-economy debate). This trade-off was preferred for the present design because the within-thread matched-control structure absorbs topic-level variation regardless of whether a thread is core or adjacent economics.

3.2 CLASSIFIER TRAINING DATA: ANNOTATION

3.2.1 Annotated Subsample and Schema

A separate subsample is used for classifier development. Eleven threads were sampled at random from the broader CMV corpus, without restriction to any topical domain, so that the classifier learns linguistic markers of construal rather than topic-specific vocabulary. Annotation is performed at the sentence level, with each task presenting a complete comment so that referential and rhetorical ambiguities can be resolved through context. Each sentence is first marked as *salient*, that is, expressing a claim, evidence, or reasoning, or non-salient. For each salient sentence, annotators assign a four-dimensional distance vector $[T, S, \mathrm{Soc}, H]$ over the Temporal, Spatial, Social, and Hypothetical dimensions. Each component takes a value in $\{0, 1, -1\}$, where 0 marks a near (concrete) cue, 1 a far (abstract) cue, and -1 the absence of any cue for that dimension; because the dimensions are coded independently, several cues can coexist in one sentence (Bar-Anan et al., 2007). Operational boundaries follow established CLT conventions (Trope and Liberman, 2010; Semin and Smith, 1999); full definitions and borderline-case rules are recorded in the annotator guidelines (Appendix A).

3.2.2 Annotators and Procedure

Three annotators contributed: the author and two peer thesis students working on the same corpus. All annotation was carried out in Label Studio,[1] with initial predictions produced by GPT-5 nano (prompted with the annotator guidelines) that annotators independently reviewed and corrected; because every label in the final dataset is human-verified, the language model functions only as a labelling aid. Annotation proceeded in two rounds. The first round served as calibration on 15 comments (83 sentences). Agreement was lowest on the Temporal dimension and moderate on Social, while Spatial and Hypothetical were already in the substantial range (Table 1). Inspection of the disagreements showed a systematic, rule-fixable boundary confusion on the Temporal and Hypothetical dimensions, so explicit boundary definitions and "split rules" for those two were added to both the written guidelines and the GPT-5 nano prompt, and the revised prompt (Appendix B) was used for all subsequent annotation. Social agreement, which had not been singled out for a rule change, improved in the second round chiefly as the annotators became more consistent with practice. A second round on 43 comments (265 sentences), produced under the revised materials, verified that agreement had improved (Section 3.2.3). Each annotator subsequently labelled additional comments individually under the revised guidelines to expand the training material. The final annotated dataset comprises 1,054 sentences across 11 threads, split at the thread level into train/dev/test partitions of 723/176/155 sentences (seed 42).

3.2.3 Inter-Annotator Agreement and Golden Standard

Agreement was measured with Krippendorff's α (Krippendorff, 2004): nominal for the binary salience decision, ordinal for each of the four CLT dimensions. The ordinal treatment respects the ordered structure of the value set $\{-1, 0, 1\}$, weighting disagreements between non-adjacent values (N/A vs. Far) more heavily than those between adjacent values. Table 1 reports α for both rounds.

Table 1: Inter-annotator agreement (Krippendorff's α) before and after guideline revision.

	Round 1	Round 2
Salience	0.822	0.820
Temporal	0.304	0.610
Spatial	0.813	0.894
Social	0.466	0.833
Hypothetical	0.768	0.886

Salience agreement was already in the "almost perfect" range ($\alpha = 0.822$, Landis and Koch 1977) after the first round. Agreement on the CLT dimensions improved substantially after guideline revision, with the dimensions that benefitted most being those for which the original boundary rules were least operationalisable in conversational text. The golden standard was derived by per-sentence, per-dimension majority vote; of 265 unique sentences, 254 (95.8%) were resolved cleanly, and the remaining 11 were resolved through joint discussion among the three annotators.

The annotated material is also highly imbalanced across the four dimensions, which has direct consequences for what the classifier can learn. Table 2 reports

[1] https://labelstud.io

the class distribution over the 895 salient sentences in the annotated set. Temporal and Spatial cues are rare: only 8.5% and 8.9% of salient sentences carry any near or far Temporal or Spatial cue, the rest being coded N/A. Social and Hypothetical cues, by contrast, are present in the large majority of salient sentences (83.1% and 94.5%). This sparsity is not a labelling artefact but a property of argumentation on CMV, and it is the main reason two of the four dimensions cannot be modelled downstream (Section 5.2).

Table 2: Class distribution of the four CLT dimensions over the 895 salient sentences in the annotated set. "Cue present" is the share of salient sentences coded near or far on that dimension (i.e. not N/A).

Dimension	N/A (−1)	Near (0)	Far (1)	Cue present
Temporal	818	32	44	8.5%
Spatial	813	35	45	8.9%
Social	150	142	602	83.1%
Hypothetical	48	603	243	94.5%

3.3 ANALYSIS SAMPLE CONSTRUCTION

The full economics corpus is filtered through a sequence of comment-level exclusions before any matched case-control sampling is applied. The joint effect of these exclusions on sample size is reported as a CONSORT-style flow diagram on the next page (cf. Schulz et al., 2010).

EXCLUSIONS. Excluded prior to matching are: self-awarded Δs; deleted or moderator-removed comments; AutoModerator and known bot accounts; comments below twenty tokens (too short to carry meaningful CLT cues, following Tan et al. (2016)); quote-only comments; and comments for which the classifier flags zero salient sentences across all four dimensions (since A_d, the per-dimension abstraction index defined in Section 4.1.2, is then undefined for all d). After topic filtering of the raw 65,169-thread CMV corpus to 11,142 economics threads (zero-shot DeBERTa (He et al., 2020; Laurer, 2024) hand-validated to 92% accuracy on a 50-thread sample, seed 42), 780,129 comments were pulled. Applying the exclusions above sequentially yields 598,272 comments retained for matching, of which 6,015 are Δ-awarded (a 97.1% retention of Δ awards from the raw pool). The matched case-control sampling described below is applied to this filtered population.

STRATIFICATION AND MATCHING. A stratum consists of one Δ-awarded comment and its k matched non-Δ controls drawn from the same thread; threads with multiple Δs contribute one stratum per case, and threads with zero Δs contribute nothing under conditional likelihood. The target matching ratio is 1:3, with 1:5 retained as a robustness check. Each non-Δ comment is matched to at most one Δ across the entire dataset (sampling without replacement, dataset-wide). When a thread contains fewer than $3 \times n_\Delta$ eligible controls, k is reduced to the largest integer for which $n_{\text{non-}\Delta} \geq k \times n_\Delta$; threads with $k = 0$ are excluded. The number of threads, strata, and total comments retained, the realised distribution of k, and the proportion of strata triggering the variable-matching fallback are reported as sample diagnostics in Section 5.4 (cf. Breslow and Day, 1980).

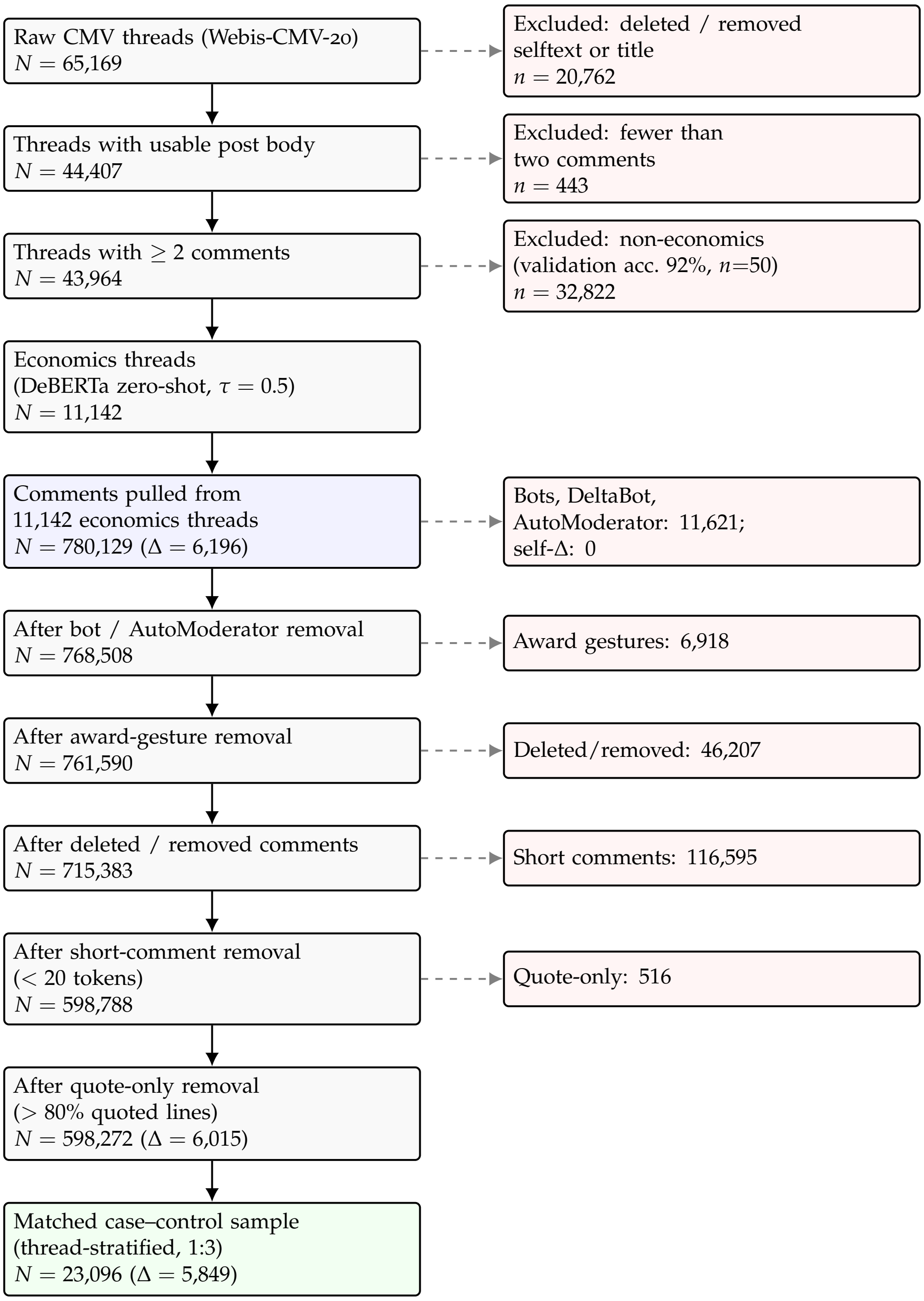

Figure 1: CONSORT-style flow diagram of the corpus construction pipeline. Thread-level filtering reduces the raw Webis-CMV-20 corpus to the economics sub-corpus; comment-level exclusions are then applied in fixed order. Solid vertical arrows trace the retained sample; dashed grey arrows show the comments removed at each step. The self-Δ check removed no comments and is folded into the bot-removal box. The validation accuracy of the topic classifier is reported on a hand-coded n=50 sample (seed 42). Δ-counts indicate awarded comments retained at the indicated stage.

3.4 TEXT PROCESSING

Posts and comments are cleaned and segmented into sentences using spaCy's neural sentence tokenizer. Each sentence is passed through the classification cascade described in Chapter 4 to produce a salience decision and, where applicable, a four-dimensional distance vector. Sentence-level vectors are aggregated to the comment level to yield the four abstraction indices used as predictors.

4 METHOD

This chapter sets out how the hypotheses are tested. Section 4.1 explains how the abstraction indices and the surface controls are built from text. Section 4.2 states the hypotheses. Section 4.3 specifies the regression models, and Section 4.4 describes inference, robustness checks, and reporting.

4.1 VARIABLE CONSTRUCTION

Constructing the explanatory variables proceeds in three steps. First, a two-stage classifier is selected and trained to label sentences for psychological distance (Section 4.1.1). Second, the sentence-level predictions are aggregated into one continuous abstraction index per dimension for each comment (Section 4.1.2). Third, a set of surface linguistic and structural controls is computed per comment (Section 4.1.3), so that the regression can test whether any abstraction effect survives adjustment for them.

4.1.1 Classifier Shootout

The classification stage operationalises psychological distance from natural language. Because the four dimensional labels are only defined for sentences carrying argumentative content, the classifier is implemented as a two-stage cascade: a binary salience classifier, followed by a multi-label classifier that, conditional on a positive salience prediction, outputs the four-dimensional vector $[T, S, \mathrm{Soc}, H]$ in a single forward pass. Identifying the best cascade is a joint exercise across three theses on the same labelled dataset, each contributing candidates evaluated on identical train/dev/test splits at the thread level (seed 42; counts 723/176/155). This thesis contributes three candidates: BERT-base (Devlin et al., 2019) and RoBERTa-base (Liu et al., 2019), fine-tuned end-to-end, and a 3B-parameter LLaMA model (Meta AI, 2024) evaluated through zero-shot prompting derived from the annotator guidelines. The two peer thesis students contribute six further candidates, yielding nine in total.

For each candidate cascade, the classifier was trained on the training set and evaluated on the development set. Each candidate was additionally evaluated on the held-out test set as a generalisation diagnostic, allowing the dev–test gap to inform model selection: a large dev–test gap, indicative of overfitting or evaluation instability, counted against a candidate in favour of one with stable behaviour across splits. The selection criterion is macro-averaged F_1-score on the development set, computed independently for the salience stage and each CLT dimension. Macro-averaging is preferred because it weights the minority classes equally with the dominant N/A class, so that poor performance on the categories of theoretical interest is reflected in the score rather than averaged away. This is what makes the collapse on Temporal and Spatial visible in the development metric (Section 5.2). Krippendorff's α between predictions and the gold-standard annotations is reported as a secondary metric, comparable to the inter-annotator analysis. The winning candidate, selected on development-set macro-F_1 subject to the stability check above, was retrained on the union of training and development data (n=899) and evaluated once on the held-out test set to produce the final reported performance.

4.1.2 Inference and Abstraction Indices

Each target sentence is passed to the classifier in isolation. This eliminates the need to choose a context-window size and keeps training and inference computationally tractable, at the cost of an asymmetry with the annotation setting, in which annotators saw the full comment context.[1] Sentence-level predictions are aggregated to the comment level. For each comment j in thread i and each dimension d, the *Abstraction Index* is

$$A_{d,ij} = \frac{\sum_{s \in S_{d,ij}} \ell_{s,d}}{|S_{d,ij}|}, \quad \ell_{s,d} \in \{0,1\}, \quad S_{d,ij} = \{s : \ell_{s,d} \neq -1\},$$

yielding four continuous indices in $[0,1]$, where 0 is fully concrete (Near) and 1 fully abstract (Far) on that dimension.

4.1.3 Computational Controls

Nine control variables are computed per comment, in two categories. *Linguistic covariates* comprise the noun-to-verb ratio, the type–token ratio, the Flesch–Kincaid grade level, and hedge-word density (Semin and Smith, 1999). The Flesch–Kincaid grade level indexes syntactic and lexical complexity, and hedge-word density (the share of tokens such as *might*, *perhaps*, or *arguably*) captures epistemic tentativeness, both of which plausibly covary with construal and with persuasive success. *Structural metrics* comprise mean sentence length, mean word length, punctuation density, paragraph count, and the number of URLs in the comment, capturing the effort and formatting features identified by Althoff et al. (2014) and Tan et al. (2016); the URL count is included because hyperlinked evidence is among the strongest correlates of Δ-success reported by Althoff et al. (2014). Descriptive statistics for all nine controls are reported in Appendix C.5. [2]

4.1.4 Dimensional Coverage and the Reduced Analysis Plan

The selected cascade (BERT-base) was diagnostically evaluated on the held-out test set on a per-dimension basis (reported in Section 5.2). The Temporal and Spatial heads collapsed to N/A predictions for nearly all sentences (per-class $F_1 \approx 0$ for Near and Far; Krippendorff $\alpha \approx 0.03$ and 0.20 respectively), reflecting the severe class sparsity already visible in the annotated data (Near and Far each below 9% of salient sentences on both dimensions). The Social and Hypothetical heads produced a usable signal, with positive (if modest) agreement against the gold standard ($\alpha = 0.14$ and 0.33). Because the regression cannot identify β_T or β_S when the classifier emits a near-constant signal on these dimensions, the primary regression analysis is restricted to Social and Hypothetical. Inference is still run on all four dimensions, and the Temporal and Spatial distributions are reported in an appendix for transparency; H1 (Temporal) and H4 (Spatial) are accordingly downgraded from primary to exploratory tests (Tier 2).

[1] A single pilot on RoBERTa Stage 2 compared the single-sentence input against a ±2-sentence context window: mean macro-F_1 was 0.43 for single sentences and 0.31 with the window. This is one check on one model in one configuration, so it is weak evidence rather than a general result; it was treated as a reason to retain the simpler single-sentence input, not as a finding about context in its own right.

[2] Two additional preprocessing steps were applied before estimation. First, each continuous control was tested for normality (Jarque–Bera); all nine are flagged ($p < 0.05$) and enter the regression under an $\ln(x+1)$ transform to reduce the influence of high-leverage observations. Second, the control set was screened for multicollinearity before estimation: three highly collinear length variables (raw token count, word-token count, and sentence count, all with variance inflation factors above 10) were dropped, leaving the nine controls listed here. The full VIF screen is reported in Appendix C.7.

4.2 HYPOTHESES

The following hypotheses are pre-committed before any regression model is fitted; analyses outside these are reported separately as exploratory. Hypotheses are presented in two tiers. Tier 1 hypotheses (H2, H3) are tested by the primary regression on the Social and Hypothetical dimensions. Tier 2 hypotheses (H1, H4) are restricted to exploratory status because the classifier collapsed on the underlying dimensions (see Section 5.2); they are reported for completeness but do not enter the confirmatory analysis.

TIER 1: PRIMARY, PRE-COMMITTED.

- **H2 (Hypothetical):** $\beta_H > 0$. On the classical, context-free reading of CLT, far (abstract) hypothetical framing increases Δ-probability (Trope and Liberman, 2010).
- **H3 (Social):** $\beta_{\text{Soc}} < 0$. Near social framing, anchored on identifiable individuals, increases Δ-probability (Small and Loewenstein, 2003).
- **H5 (Surface-Feature Robustness):** The Tier 1 β_d coefficients retain their sign and remain distinguishable from zero after surface covariates (length, lexical diversity, hedging, complexity) are entered as adjustment covariates. That is, the abstraction–Δ link is not fully accounted for by the surface features identified in prior CMV research (Althoff et al., 2014; Tan et al., 2016). H5 is tested by comparing an unadjusted specification (abstraction indices only, M0) against the fully adjusted Model 1 (M1): for each $d \in \{\text{Soc}, H\}$, H5 holds if $\text{sign}(\beta_d^{M1}) = \text{sign}(\beta_d^{M0})$ and β_d^{M1} remains distinguishable from zero.

Both pre-committed Tier 1 predictions follow established accounts: H2 the classical, main-effect reading of CLT (Trope and Liberman, 2010), and H3 the identifiable-victim effect (Small and Loewenstein, 2003). The fit account reviewed in Section 2.1.2 predicts instead that concrete framing dominates whenever the audience is reasoning about something psychologically near, which in the present setting would imply a concrete advantage on *both* dimensions. For H2 that is the opposite of the pre-committed prediction; for H3 it is a second rationale converging on the same direction. That fit prediction was not part of the pre-committed analysis plan; it is tested in a post-hoc analysis (Section 4.3.3, with results in Section 5.8), and is noted here only to fix what was and was not pre-committed.

TIER 2: EXPLORATORY, REPORTED BUT NOT CONFIRMATORY.

- **H1 (Temporal, exploratory):** Originally $\beta_T > 0$ (Liberman and Trope, 1998; Fujita et al., 2006); downgraded to exploratory because the Temporal head of the selected cascade collapsed to N/A predictions on the held-out test set (Section 5.2). The regression cannot identify β_T when its input is near-constant. Temporal predictions are nonetheless reported descriptively in an appendix.
- **H4 (Spatial, exploratory):** Originally $\beta_S \neq 0$, direction unspecified, given mixed prior evidence (Trope and Liberman, 2010); downgraded to exploratory on the same grounds as H1. Spatial predictions are reported descriptively in an appendix.

4.3 MODEL SPECIFICATIONS

4.3.1 Model 1: Conditional Logistic Regression

The primary model is a conditional logistic regression (Breslow and Day, 1980) estimated within matched strata:

$$\operatorname{logit} P(\Delta_{ij} = 1 \mid \text{stratum } i) = \sum_d \beta_d A_{d,ij} + \gamma Z_{ij}.$$

Stratum-specific intercepts drop from the conditional likelihood, identifying the β_d from within-stratum variation alone and absorbing all time-invariant thread-level confounding non-parametrically. H1–H4 are tested by the signs and significance of the β_d coefficients.

A causal mediation decomposition (NDE/NIE; Imai et al., 2010) was considered as an alternative and rejected. The reason is identification: the decomposition recovers the natural direct and indirect effects of A_d only under the sequential ignorability assumption (VanderWeele, 2015), which requires no unmeasured confounder of mediator and outcome given treatment. In an observational subreddit corpus, plausible confounders of both the surface mediators and the Δ outcome that vary *within* threads (e.g., author skill, topical familiarity) are unmeasured and cannot be controlled away by the stratum design. Entering the surface features as adjustment covariates within Model 1 addresses the empirical question of whether the abstraction–Δ link survives surface adjustment without resting on an identifying assumption the data cannot support.

4.3.2 Robustness Model: Mixed-Effects Logistic Regression

Model 1 absorbs thread-level heterogeneity but is silent on commenter-level clustering. A generalised linear mixed model with crossed thread and author random intercepts is therefore estimated as a structurally distinct robustness specification (Bates et al., 2015):

$$\operatorname{logit} P(\Delta_{ij} = 1) = \beta_0 + \sum_d \beta_d A_{d,ij} + \gamma Z_{ij} + u_{\text{thread}(i)} + u_{\text{author}(j)}.$$

Agreement with Model 1 on the directions and approximate magnitudes of β_d is taken as evidence that the conclusions are not artefacts of the matched design.

4.3.3 Post-hoc Analysis: Construal Fit

The models above test whether a comment's own construal level predicts persuasion. A separate prediction follows from the fit account introduced in Section 2.1.2: persuasion should be greatest when the construal level of an argument matches that of the audience it addresses (Kim et al., 2009; White et al., 2011). This was not part of the pre-committed plan and is reported as exploratory, with no directional prediction registered in advance. To test it, the submission text of each thread, the original poster's stated view, was passed through the same two-stage cascade and aggregated by the procedure of Section 4.1.2, yielding thread-level indices A^{OP}_{Soc} and A^{OP}_{H}.

Because the original poster's construal is constant within a thread, it is absorbed by the stratum-specific intercepts and cannot enter the conditional model as a main effect. Fit is therefore tested through two specifications. The interaction specification (F2) adds the products of comment and OP construal, for which the fit account predicts positive coefficients:

$$\operatorname{logit} P(\Delta_{ij} = 1 \mid \text{stratum } i) = \sum_{d \in \{\text{Soc}, H\}} \beta_d A_{d,ij} + \sum_d \theta_d \, A_{d,ij} \, A_{d,i}^{OP} + \gamma Z_{ij}.$$

The distance specification (F3) replaces the products with the absolute construal gap, for which fit predicts negative coefficients, a smaller gap meaning a closer match and higher Δ-odds:

$$\operatorname{logit} P(\Delta_{ij} = 1 \mid \text{stratum } i) = \sum_{d \in \{\text{Soc}, H\}} \beta_d A_{d,ij} + \sum_d \delta_d \left| A_{d,ij} - A_{d,i}^{OP} \right| + \gamma Z_{ij}.$$

Both are estimated at the 1:3 and 1:5 matching ratios, with the Model 1 controls and thread-clustered standard errors. An OP-main-effects specification (F1) is reported in Appendix C.9 for completeness but, for the reason just given, is not identified.

4.4 INFERENCE, ROBUSTNESS, AND REPORTING

Standard errors for Model 1 are clustered at the thread level to account for residual correlation between strata sharing a thread. Two further robustness checks are run: re-estimation of Model 1 at a 1:5 matching ratio, and re-estimation excluding strata for which the variable-matching fallback was triggered. All hypothesis tests are two-tailed at $\alpha = 0.05$; 95% confidence intervals are reported throughout. Coefficients are reported as log-odds with thread-clustered robust standard errors and as odds ratios over the full $[0, 1]$ range of each abstraction index. Absolute risk differences are not reported: the conditional likelihood eliminates the stratum intercepts, so the matched case–control design does not identify baseline probabilities. Effect magnitudes rather than p-values carry the substantive argument (cf. Ziliak and McCloskey, 2008), and effect directions are interpreted against the pre-committed hypotheses H1–H5.

5 RESULTS

5.1 CLASSIFIER SHOOTOUT

Table 3 reports development-set macro-F_1 for the three candidate cascades contributed by this thesis, together with the held-out test-set macro-F_1 used as a generalisation diagnostic. Selection was on development-set performance; test numbers informed only the dev–test stability check fixed in the protocol. Six further candidates were contributed by peer thesis students working on the same labelled dataset under the same shared protocol and are reported in their respective theses (Wegink, 2026; Jans, 2026); none exceeded the BERT-base candidate on overall development-set macro-F_1 (averaged across the salience stage and the four CLT dimensions).

Table 3: Dev-set and held-out-set macro-F_1 for the three cascade candidates contributed by this thesis. Stage 1 is binary salience ($n_{\text{dev}} = 176$, $n_{\text{test}} = 155$); Stage 2 is the four-dimensional CLT vector evaluated on gold-salient sentences ($n_{\text{dev}} = 147$, $n_{\text{test}} = 127$). Selection was on dev; test numbers are generalisation diagnostics.

Candidate	Dev S1	Dev S2 mean	Test S1	Test S2 mean
BERT-base (Devlin et al., 2019)	0.874	0.450	0.852	0.423
RoBERTa-base (Liu et al., 2019)	0.840	0.434	0.843	0.469
LLaMA-3.2-3B (Meta AI, 2024)	0.455	0.256	—	—

BERT-base is the winning cascade on the pre-committed criterion: the highest overall development-set macro-F_1, averaged across the salience stage and the four CLT dimensions (Stage 1 $F_1 = 0.874$, Stage 2 mean $F_1 = 0.450$). RoBERTa-base scores marginally higher on the test mean (0.469 vs. 0.423) but lower on the dev mean, a dev–test divergence indicative of evaluation instability under a single small test fold; selecting on a metric the dev split was held out to inform was the protocol fixed before the test set was touched. On the salience stage both encoders agree closely with the gold labels (Stage 1 Krippendorff $\alpha = 0.71$ for BERT and 0.69 for RoBERTa on the held-out set), in sharp contrast to the prompted LLaMA cascade ($\alpha = -0.09$), discussed next. The BERT-base parameter count (∼110M) is also lower than that of the larger peer candidates, materially affecting wall-clock cost for sentence-level inference over the 598,272-comment corpus.

The prompted LLaMA-3.2-3B cascade is reported but not competitive. Stage 1 collapses to predicting salient for every input (macro-$F_1 = 0.455$; Krippendorff $\alpha = -0.09$, worse than chance), and Stage 2 mean macro-F_1 is 0.256 with negative α on three of four dimensions, indicating systematic class-direction errors rather than random noise. The theoretical reading of this negative result is given in Section 6.2.[1]

The selected BERT-base cascade was retrained on the union of training and development data ($n = 899$) and evaluated once on the held-out test set ($n = 155$). Final reported performance is Stage 1 macro-$F_1 = 0.869$ (accuracy 0.923) and Stage 2 mean macro-$F_1 = 0.436$ (the unweighted mean of the four per-dimension macro-F_1 scores in Table 4). The per-dimension breakdown of Stage 2 is the subject of Section 5.2.

[1] Two implementation notes for the LLaMA pipeline. (i) The calibration step was intended to operate on a 30-sentence subset of dev but selected unique task IDs rather than (task, sentence) pairs; because dev contains only 24 unique tasks, calibration ran on the full 176-sentence split. The prompt was not iterated against this output, decoding is greedy and therefore deterministic, and the reported score is the committed evaluation. (ii) The Llama-3 chat template auto-prepends a fixed knowledge-cutoff date and the current date to the system prompt; this is reproducible across runs.

5.2 DIMENSIONAL COVERAGE OF THE SELECTED CASCADE

Aggregate Stage 2 performance hides systematic variation across the four CLT dimensions. Table 4 reports per-dimension diagnostics for the final retrained BERT-base cascade on the held-out test set.

Table 4: Per-dimension performance of the retrained BERT-base cascade on the held-out test set ($n = 127$ salient sentences). Per-class F_1 is given for N/A, Near, and Far; a near-zero F_1 on Near and Far indicates a head that predicts N/A for nearly every sentence.

Dimension	Macro-F_1	Kripp. α	Per-class F_1 (N/A, Near, Far)	Status
Temporal	0.308	0.027	(0.92, 0.00, 0.00)	Collapsed
Spatial	0.397	0.196	(0.91, 0.00, 0.29)	Collapsed
Social	0.448	0.143	(0.06, 0.51, 0.77)	Usable
Hypothetical	0.590	0.333	(0.56, 0.79, 0.42)	Usable

The Temporal and Spatial heads collapsed to near-constant N/A predictions. The Temporal head produces zero per-class F_1 on both the Near and the Far category; the Spatial head does the same on Near and reaches only 0.29 on Far. Krippendorff α between predictions and the gold standard confirms the failure: 0.03 on Temporal and 0.20 on Spatial, in both cases at or below the threshold conventionally treated as minimally usable agreement (Krippendorff, 2004). This pattern follows directly from the class imbalance rather than from any implementation fault: Temporal Near and Far together make up 9.5% of salient training sentences and Spatial Near and Far only 7.4%, so a classifier minimising expected loss has little incentive to predict either minority class. The Temporal collapse recurs in every candidate cascade in the shootout; Spatial recovery is weak throughout, although the runner-up RoBERTa cascade does recover the Spatial dimension more than BERT (per-dimension figures in Appendix C.6). Because the conditional logistic regression cannot identify β_T or β_S when its input feature is near-constant across observations, the Temporal and Spatial dimensions are excluded from the confirmatory analysis. This is the basis for the Tier 1 (Social, Hypothetical) versus Tier 2 (Temporal, Spatial) hypothesis split fixed in the Methods chapter (Section 4.2).

The Social and Hypothetical heads produce a usable signal. On the Social head the classifier reaches $F_1 = 0.77$ on the Far class and $F_1 = 0.51$ on Near, with almost all of the residual error on the N/A class ($F_1 = 0.06$). The three-class Krippendorff $\alpha = 0.14$ is low, but it is pulled down by N/A confusion rather than by errors on the Near–Far distinction the regression actually uses. The abstraction index is computed only over sentences the classifier assigns a cue (predicted Near or Far); restricted to those sentences, the ones that enter the index, the predicted Near/Far label matches the gold label in 83% of cases (80 of 96, from the test-set confusion matrix). The contrast that drives the regression is therefore recovered considerably better than the three-class α suggests. The same matrix shows the head's limit, however: of the 126 gold-cue Social sentences, 30 (24%) are predicted N/A and dropped from the index, which attenuates the index toward the mean and is reflected in the conservative effect sizes noted in Section 6.3.

5.3 CONVERGENCE OF THE ABSTRACTION INDICES

Construal Level Theory holds that the four distance dimensions are cognitively interrelated and converge on a common level of abstraction (Bar-Anan et al., 2007). Were that so in this corpus, the four comment-level indices would be positively correlated. Computed over the full inference corpus with pairwise-complete observations, they are not. The two indices that enter the confirmatory analysis, A_{Soc}

and A_H, correlate at $r = -0.28$ ($n = 543{,}740$): socially near comments tend toward hypothetically far framing rather than toward a shared concrete pole. A principal-component analysis of the two reproduces this. Its first component is a contrast, with A_{Soc} loading $+0.71$ and A_H loading -0.71, and it accounts for 64% of the variance, where a single shared factor would have loaded both with the same sign. The remaining pairs involve the collapsed Temporal and Spatial indices, are weak, and are defined for a minority of comments; the full matrix is given in Appendix C.8. On these two dimensions the indices behave as separable axes, not as one underlying construal level.

5.4 SAMPLE CONSTRUCTION DIAGNOSTICS

The CONSORT-style funnel of corpus construction is given as Figure 1 in the Data and Material chapter, which traces the reduction from the 65,169 raw threads to the 598,272-comment inference corpus (6,015 Δ-awarded, a 97.1% retention of awards through the exclusion sequence). The diagnostics that figure does not show are reported here.

The matched case–control sample was drawn from the 598,272-comment inference corpus by within-thread risk-set matching (Section 5.5). Of the 5,881 comments eligible as cases (both A_{Soc} and A_H defined), 5,849 were retained; 32 were dropped, five lying in two threads with no eligible controls and 27 at a zero-control floor. At the primary 1:3 ratio this yields 23,096 rows (5,849 cases, 17,247 controls) across 5,849 strata, drawn from 3,556 threads contributing at least one case (seed 42). The CONSORT thread funnel is 11,142 (topic filter) $\rightarrow$ 11,134 (scored) $\rightarrow$ 11,127 (≥ 1 eligible comment) $\rightarrow$ 3,556 (≥ 1 eligible case).

Table 5 reports the realised matching ratio. The full $k = 3$ ratio was achieved for 5,626 strata (96.2%); the variable matching fallback was triggered for 223 strata (3.8%), which matched at $k = 1$ or $k = 2$ for want of eligible within-thread controls. The 1:5 robustness sample (seed 43) comprises 33,890 rows and exhibits a higher fallback rate (545 strata, 9.3%), as expected when a larger control demand strains the within-thread supply. The matched sample spans 8,868 unique authors at 1:3 and 11,235 at 1:5.

Table 5: Realised per-stratum control count (k) for the primary 1:3 and robustness 1:5 matched samples. Strata at k below the target ratio reflect the variable-matching fallback.

Sample	k=1	k=2	k=3	k=4	k=5
1:3 primary	77	146	5,626	—	—
1:5 robustness	77	146	136	186	5,304

5.5 PRIMARY ANALYSIS: CONDITIONAL LOGISTIC REGRESSION

Table 6 reports the primary conditional logistic regression in two nested specifications: an unadjusted model with the two abstraction indices alone (M0) and the pre-committed adjusted model adding the nine surface controls (M1, Section 5.5). Both use standard errors clustered at the thread level.

The two exposures behave differently under adjustment. The Social index reverses sign. Unadjusted, higher social abstraction is associated with *greater* Δ-odds (OR 1.14, $p = .005$); adjusted for surface features, the association reverses to *lower* Δ-odds (OR 0.86, $p = .015$). This is a suppression pattern: the length and complexity controls, which correlate with both the social index and Δ-receipt, mask a

Table 6: Conditional logistic regression of Δ-award on the Tier 1 abstraction indices, primary 1:3 matched sample ($n = 23{,}096$; 5,849 strata). M0 is exposures only; M1 adds nine log-transformed surface controls. Estimates are log-odds with thread-clustered robust SEs; OR with 95% CI.

	Unadjusted (M0)			Adjusted (M1)		
Variable	$\hat{\beta}$ (SE)	p	OR [95%]	$\hat{\beta}$ (SE)	p	OR [95%]
A_{Soc}	0.134 (0.048)	.005	1.14 [1.04, 1.25]	−0.155 (0.064)	.015	0.86 [0.76, 0.97]
A_H	−0.454 (0.048)	$< .001$	0.63 [0.58, 0.70]	−0.108 (0.067)	.105	0.90 [0.79, 1.02]

Concordance: M0 = 0.533, M1 = 0.727. Surface controls reported in Appendix C.4.

negative within-stratum association. Net of them, concrete (Near) social framing predicts Δ-success, the direction predicted by H3, but the H3-consistent estimate is conditional on adjustment and its causal reading is discussed in Section 6.1.

The Hypothetical index is negative in both specifications, attenuating from OR 0.63 ($p < .001$) to OR 0.90 ($p = .105$) once surface features enter. Its direction is the opposite of the positive coefficient predicted by H2 throughout, and most pronounced before adjustment; H2 is not supported. Significance of the adjusted estimate is recovered in the better-powered 1:5 sample (Section 5.6).

Discrimination is carried almost entirely by the surface controls: exposures-only concordance is 0.533, barely above chance, rising to 0.727 once controls are added, i.e. the adjusted model correctly ranks a Δ-awarded comment above a non-awarded one roughly 73% of the time, against the 50% of chance. The abstraction effects are robust in direction but modest in magnitude relative to surface characteristics.

H5 asks whether the Tier 1 coefficients retain their sign and remain distinguishable from zero once surface features are entered. The two dimensions diverge and H5 is not a single pass/fail. A_H retains its negative sign across M0 and M1 but is distinguishable from zero only in the 1:5 sample. A_{Soc} does not retain its sign: the surface features suppress rather than inflate its effect. The abstraction–Δ association is not reducible to surface features for either dimension, but the literal sign-stability criterion of H5 holds only for A_H.

5.6 ROBUSTNESS CHECKS

Table 7 reports the Tier 1 estimates across six specifications: the primary 1:3 conditional logit (M1) and GLMM (M3), a fallback-excluded re-fit retaining only full-k=3 strata (n=22,504; 5,626 strata), and both models refitted on the 1:5 matched sample (n=33,890). All six yield negative estimates for both exposures.

Table 7: Odds ratios for the two Tier 1 abstraction indices across six specifications. All models include nine log-transformed surface controls except M0 (reported in Table 6 for reference). M1 uses cluster-robust SEs at thread level; M3 uses model-based SEs with random effects $(1|\textit{thread}) + (1|\textit{author})$.

Model	Sample	A_{Soc} OR (p)	A_H OR (p)
M1 clogit	1:3 primary (n=23,096)	0.856 (.015)	0.898 (.105)
M3 GLMM	1:3 primary (n=23,096)	0.760 ($<$ .001)	0.926 (.228)
M1 clogit	1:3 fallback-excl. (n=22,504)	0.861 (.021)	0.905 (.140)
M3 GLMM	1:3 fallback-excl. (n=22,504)	0.774 ($<$ .001)	0.927 (.244)
M1 clogit	1:5 robustness (n=33,890)	0.860 (.011)	0.862 (.015)
M3 GLMM	1:5 robustness (n=33,890)[a]	0.727 ($<$ .001)	0.869 (.022)

[a] `bobyqa` exceeded the time limit at 11,235 author levels; `Nelder_Mead` was used and converged.

A_{Soc} is negative and significant in all six specifications (OR 0.727–0.861; all $p \leq .021$): the direction and significance of the Social effect are robust to fallback-stratum inclusion, matching ratio, and model class. A_H is consistently negative but significance is sample-size dependent: not significant in any 1:3 specification ($p \geq .105$), significant in both 1:5 specifications (M1 OR 0.862, $p = .015$; M3 OR

0.869, $p = .022$). The 1:3 primary was underpowered for the smaller A_H effect; the 1:5 sample resolves H2 in the refuted direction.

In all three M3 specifications the thread random-effect variance is negligible ($\approx 8.7 \times 10^{-8}$), confirming that within-thread matching absorbed thread-level confounding. Author random-effect variance is stable at ≈ 0.39 across M3 specs, indicating between-author heterogeneity in Δ-receipt that substantially exceeds the abstraction effects in magnitude.

5.7 EXPLORATORY RESULTS ON TEMPORAL AND SPATIAL

Table 9 (Appendix C.3) reports the full-corpus availability and distribution of the Temporal and Spatial indices. Both are defined for a minority of comments (38.2% and 25.7% respectively), reflecting the sparse non-N/A sentence-level predictions of the collapsed heads aggregated to the comment level (Section 5.2). Among comments where they are defined, neither index separates Δ-awarded from non-awarded comments (Temporal: 0.665 vs. 0.653; Spatial: 0.511 vs. 0.502). The defined values carry no detectable signal; consistent with the pre-committed plan, no inferential test is performed and H1 and H4 remain exploratory.

5.8 EXPLORATORY RESULTS: CONSTRUAL FIT

Table 15 (Appendix C.9) reports the fit specifications of Section 4.3.3. The original-poster indices are abstract on the social dimension and factual on the hypothetical one (A^{OP}_{Soc} mean 0.72, A^{OP}_{H} mean 0.34), consistent with economics submissions that argue about groups and institutions while stating the view as fact. As anticipated, the OP main effects (F1) are absorbed by the strata and not estimable.

The distance specification (F3) supports the fit prediction on both estimable dimensions and replicates across matching ratios. A larger comment–OP construal gap lowers Δ-odds on the social dimension (OR 0.64, $p < .001$ at 1:3; OR 0.68, $p < .001$ at 1:5) and on the hypothetical dimension (OR 0.79, at $p = .015$ and $p = .008$ respectively). The interaction specification (F2) corroborates the social result, where the comment–OP product is positive (OR 1.91, $p = .002$, replicated at 1:5), but not the hypothetical one, where the product form is non-significant in both samples. The distance form is the more direct test of alignment, and it is the one that holds throughout.

The two dimensions differ in their relation to the concreteness effect of Section 5.5. On the social dimension the comment-level coefficient and the alignment term are jointly significant. On the hypothetical dimension the comment-level A_H effect, significant on its own in the better-powered 1:5 sample, falls to non-significance once alignment is added (OR 0.91, $p = .14$) while alignment remains significant. The two readings of this pattern are taken up in Section 6.1.

6 DISCUSSION AND CONCLUSION

6.1 INTERPRETATION OF PRIMARY HYPOTHESES

Both Tier 1 estimates are negative across every adjusted specification, indicating that concrete construal is associated with greater Δ-odds in this setting. On a context-free reading of CLT, on which abstract framing promotes attitude change (Trope and Liberman, 2010), this is the wrong direction. It is, however, the direction predicted by the fit account of CLT-based persuasion (Kim et al., 2009; White et al., 2011): the ChangeMyView addressee is an identified individual deliberating a specific, present claim, a psychologically near situation, and the matching principle holds that near framing persuades most when the audience is reasoning about something near. The uniform concrete advantage found here is therefore better read as consistent with the fit refinement of CLT than as a flat refutation of it. The design cannot establish fit as the mechanism, since it does not manipulate the addressee's distance, but the result aligns with what that mechanism predicts, and a post-hoc test of the matching prediction (Section 5.8), discussed at the end of this section, bears it out. A simpler reading should be kept in view throughout: concrete, specific language is also easier to process, and ease of processing can raise persuasion on its own, independently of construal (Alter and Oppenheimer, 2009). The two estimable hypotheses are discussed in turn below.

H3 (Social) receives only qualified support. In the adjusted model the sign is H3-consistent: net of surface controls, concrete social framing is associated with higher Δ-odds (OR 0.856, $p = .015$), and this holds across all six robustness specifications. The unadjusted estimate, however, points the other way (OR 1.14, $p = .005$), so the H3-consistent direction is not robust to specification, it holds only when the surface controls are treated as confounders to be adjusted away, and reverses when they are not. This links H3 directly to H5: because H5 requires the Tier 1 effects to keep their sign under adjustment, the Social effect satisfies H3 only in the adjusted model while failing H5's sign-stability criterion. The defensible reading is that the Social result is directionally consistent with a concrete-construal advantage but fragile and adjustment-dependent, not a robust main effect. Where the sign does hold, near-level social anchoring, locating arguments in identifiable individuals rather than statistical aggregates (Small and Loewenstein, 2003), is associated with greater persuasive success once length and complexity are held constant.

H2 (Hypothetical) is refuted, and by the effect-size standard adopted in Section 4.4 the refutation is clear-cut. The Hypothetical coefficient is negative in all six specifications with a stable magnitude (OR between roughly 0.86 and 0.90); the 1:5 sample, which has the most power, additionally returns it as significant (OR 0.862, $p = .015$). The consistent negative magnitude, not the single significant p-value, is the basis for the verdict, with the 1:5 result confirming rather than establishing the pattern. Abstract hypothetical framing reduces Δ-odds throughout, the opposite of the CLT prediction.

This refutation must be read against a construct-validity concern specific to the Hypothetical dimension. As operationalised here (Appendix A), a near Hypothetical cue is a factual or experiential assertion ("data shows", "I saw"), while a far cue is speculation or a counterfactual ("suppose", "if X then Y"). A negative Hypothetical coefficient therefore says that factual assertion outperforms speculation. That is consistent with CLT, but it is equally what a simpler epistemic-stance or credibility account predicts, where confident, evidence-grounded claims persuade more than hedged speculation, and that account has its own standing independent of construal. The concern is sharpened by the fact that hedge-word density, one of the surface controls, taps the same fact-versus- speculation axis, so the construct and one of its adjustment covariates partly overlap. The Hypothetical result is thus better described as evidence that concrete, factual framing persuades than as a clean test of psychological distance: consistent with CLT, but not isolating a mechanism that CLT uniquely predicts. The Social dimension, whose near/far contrast is between identifiable individuals and abstract collectives, is less exposed to this reading, which is a further reason to weight it more heavily in any theoretical conclusion. For Hypothetical, then, the fit account and a simpler credibility account make the same directional prediction and cannot be separated here; the Social dimension, free of that confound, is the cleaner evidence for a construal-fit reading.

H5 is met for A_H (sign stable across M0 and M1, distinguishable from zero in the 1:5 sample) but not for A_{Soc}, whose sign reverses under adjustment. The abstraction–Δ association is not reducible to surface features for either dimension, but the mechanism differs: suppression for Social, attenuation for Hypothetical. Classifier measurement error on both dimensions attenuates $\hat{\beta}_d$ toward zero (Section 6.3); reported effect sizes are conservative.

The fit account, tested directly. The reading above treats fit as an interpretive lens on a comment-level result. A post-hoc analysis tested its distinctive prediction instead (Section 5.8). Persuasive success tracks the construal distance between a comment and the original poster it answers: on both estimable dimensions a closer match raises Δ-odds, and the effect holds at both matching ratios. This sharpens the two hypotheses. On the social dimension, alignment with the addressee adds to the concrete-construal advantage rather than replacing it. On the hypothetical dimension, alignment largely is the effect: the apparent advantage for factual framing is, more precisely, an advantage for matching the original poster's own factual stance. The pattern is the one the fit account predicts, observed directly rather than inferred. The earlier caveat returns at the level of the dyad, since matching an addressee's framing is also what a credibility or responsiveness account would call engagement; the design supports the fit prediction without isolating fit as the mechanism, and the test is correlational and exploratory.

6.2 THE CLASSIFIER COLLAPSE AND ITS THEORETICAL IMPLICATIONS

The collapse of the Temporal and Spatial heads is the most visible finding of the cascade evaluation, and it has a direct consequence for the hypotheses: H1 (Temporal) and H4 (Spatial) cannot be tested at all. A regression coefficient is only identified when its predictor varies across observations, and a head that returns N/A for almost every sentence yields a near-constant index. The two Tier 2 hypotheses are therefore left open, not refuted, by this study. The collapse itself, however, is informative, and two readings are plausible.

The first is operational: at 9.5% Temporal and 7.4% Spatial non-N/A prevalence in the annotated training data, no candidate in the shootout recovered the minority classes, and a more permissive metric would have masked the failure rather than revealed it. Under this reading the conclusion is about annotation supply: more data with deliberate over-sampling of the minority categories would likely produce usable heads. The second reading is theoretical and more cautious. The Temporal and Spatial dimensions, as operationalised in standard CLT instruments (Trope and Liberman, 2010; Semin and Smith, 1999), may simply be less linguistically marked in argumentative discourse on policy questions than the Social and Hypothetical dimensions are. CMV economics threads concern groups, institutions, and counterfactual policy scenarios; explicit temporal and spatial anchoring is infrequent because the underlying arguments are pitched at the level of social aggregates and hypotheticals. On this reading, the skew in the annotated distribution reflects how these arguments are actually framed, rather than an artefact of which threads were sampled. The present data cannot adjudicate between the two readings, but the second carries a substantive implication: psychological distance is not uniformly recoverable from argumentative text. Social and hypothetical distance are lexicalised in policy debate while temporal and spatial distance largely are not, which bounds what CLT instruments can be expected to measure in this register (Semin and Smith, 1999; Zymla et al., 2024).

A related observation comes from the dimensions that did survive. Section 5.3 found the social and hypothetical indices weakly *negatively* correlated rather than positively, with no single component summarising them. Construal Level Theory treats the distance dimensions as facets of one underlying construal level (Bar-Anan et al., 2007); in this corpus they behave as separable axes. The collapse of Temporal and Spatial and the non-convergence of Social and Hypothetical point the same way: the unified construal continuum that holds in laboratory tasks does not reproduce cleanly in argumentative policy text, where the dimensions are unevenly marked and only loosely coupled.

The prompted-LLM result is a negative finding worth stating on its own terms. The 3B LLaMA cascade failed differently from the encoder candidates: Stage 1 collapsed to a single class regardless of input, and Stage 2 over-predicted Far on three of four dimensions, tracking the label distribution of the few-shot examples. This is consistent with the few-shot prior dominating the input signal at this model scale and example budget. The finding speaks to prompted small instruction-tuned LLMs as a substitute for fine-tuned encoders on narrow annotation tasks; it does not extend to larger models, which were not tested here.

One further point bears on the collapse. The selection criterion (Section 4.1.1) chose BERT-base on its higher average development macro-F_1, a margin of 0.450 versus 0.434 over RoBERTa-base that is well within fold noise on 147 development sentences. On the held-out test set the discarded RoBERTa cascade recovers the *far* Spatial class far better than BERT (per-class $F_1 = 0.59$ versus 0.29; overall Spatial $\alpha = 0.51$ versus 0.20; Appendix C.6), correctly labelling 10 of 17 gold-far spatial sentences rather than collapsing them to N/A, though the near Spatial class remains essentially unrecovered in both models, and these estimates rest on few minority-class test sentences. Optimising a single averaged score was defensible and was fixed before the test set was examined, but for a research goal that valued dimensional coverage it was arguably the wrong criterion: the model that would have delivered at least a partially usable Spatial head was the one the criterion rejected. A coverage-weighted selection rule is a reasonable alternative for future work.

6.3 LIMITATIONS

Five limitations bound the inferences. First, the annotated dataset is small ($n =$ 1,054 sentences, 11 threads) and the test split is a single small fold ($n = 155$), so all reported classifier performance is estimated with non-trivial single-fold variance.

Relatedly, the decision to classify each sentence in isolation rests on a single pilot comparison on one model (Section 4.1.2); whether a wider context window would help is not settled by that one check, and a fuller comparison across models and window sizes is left to future work. Second, identification in Model 1 is from within-thread variation only; effects that operate at the thread level (topic, framing of the original post, time of posting) are absorbed by the matched design and cannot be estimated here.

Third, classifier error on Social and Hypothetical ($\alpha = 0.14$ and 0.33) propagates into the regression as measurement error in the predictors, biasing β_d toward zero; the reported magnitudes are therefore conservative.

Fourth, the analysis is observational. Even after thread-level stratification and surface characteristic adjustment, unmeasured confounders, author skill, topical familiarity, posting time, the specific original poster being addressed remain plausible alternative explanations of any $\beta_d \neq 0$. The decision to treat surface characteristics as adjustment covariates rather than entering them in a formal mediation decomposition is itself an acknowledgement that the data do not support causal identification of the abstraction–surface-feature pathway.

Fifth, the sign reversal of A_{Soc} between the unadjusted and adjusted models raises the question of whether the nine surface controls act as confounders of, or mediators in, the abstraction–Δ pathway. If they mediate, adjusting partially blocks the causal path and the unadjusted total effect is the relevant quantity; if they confound, the adjusted M1 estimate is correct. The observational design cannot settle this, and the reported effect direction for Social is conditional on the confounder interpretation.

6.4 CONCLUSION

The thesis pursued two goals: building a validated CLT annotation cascade scalable to a large online corpus, and testing whether construal-level framing predicts persuasive success on ChangeMyView.

The cascade delivers usable classification on the Social and Hypothetical dimensions (Stage 2 macro-F_1 0.448 and 0.590) while establishing that Temporal and Spatial signal is undetectable at current annotation scale, a result that is informative in its own right, constraining what CLT instruments can be expected to recover in argumentative discourse. Prompted small instruction-tuned LLMs are not a substitute for fine-tuned encoders on narrow annotation tasks at this scale.

The regression evidence points consistently toward a concrete-construal advantage in economics and policy argumentation on ChangeMyView. Of the two testable hypotheses, H3 (Social) is supported in the adjusted model, while H2 (Hypothetical) is refuted: the Hypothetical effect is negative throughout, the opposite of the classical CLT prediction. H1 (Temporal) and H4 (Spatial) could not be tested, because the corresponding classifier heads collapsed and left their indices near-constant; they remain open rather than refuted. H5 is met in part: the abstraction–Δ association survives surface adjustment for both estimable dimensions, but only the Hypothetical effect keeps its sign across the unadjusted and adjusted models, whereas the Social effect reverses sign under adjustment and is thus conditional on reading the surface controls as confounders rather than mediators. Taken together, both estimable results show a concrete advantage that the classical, main-effect reading of CLT does not predict but the fit account does: in a psychologically near setting such as real-time CMV debate, where the addressee is an identified individual weighing a specific claim, concrete framing that engages the opponent's position outperforms abstract-principle framing. The design cannot establish fit as the mechanism, but the data align with what it predicts. The effects are modest, the design is observational, and between-author heterogeneity dwarfs the abstraction signal. The domain restriction cuts both ways: filtering to economics threads removed cross-topic confounding and made the matched design tractable, but it is also why the Temporal and Spatial dimensions could not be studied, since economics argument rarely anchors claims to specific times or places. The findings should therefore be read as scoped to policy-style argumentation rather than to ChangeMyView as a whole. Replication under a stronger causal identification strategy, and with annotation resources sufficient to recover the Temporal and Spatial dimensions, would consolidate the conclusion.

CODE AND DATA AVAILABILITY

All code for the classifier shootout, the data construction pipeline, the inference pipeline on the economics sub-corpus, and the R regression analyses is publicly available at `https://github.com/KrisHoffmann/Bachelor-Thesis-IK`. The Webis-CMV-20 corpus is publicly available at `https://zenodo.org/records/3778298` (Kolyada et al., 2020). The annotated dataset, train/dev/test splits, and BERT-base model checkpoint are archived in the repository's `v1.0` release.

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A ANNOTATOR GUIDELINES

The following guidelines were issued to all annotators (the author and two peer thesis students). The Round 1 calibration showed the lowest agreement on the Temporal dimension and moderate agreement on Social; the systematic, rule-fixable confusion was concentrated on the Temporal and Hypothetical dimensions, and the operational boundaries for those two were tightened in Round 2 in both these guidelines and the GPT-5-nano annotation-assist prompt (see Appendix B). The text below reproduces the guidelines as they were presented to annotators.

PROJECT OBJECTIVE

We are analysing how users on ChangeMyView (CMV) use different levels of mental abstraction to persuade others. Instead of labelling whole posts, we use *salient sentence annotation*. This allows us to map the "flow" of an argument between abstract principles and concrete evidence.

THE TWO-STEP ANNOTATION PROCESS

Annotators (and the language-model assistant) follow this logic for every sentence in a post.

STEP 1: SALIENCE FILTER. Identify whether the sentence is a salient sentence. A sentence is salient if it contains:

- **The claim:** the main point or opinion being argued.
- **The evidence:** facts, statistics, or personal anecdotes used to support a claim.
- **The reasoning:** the logical "because" connecting a claim to evidence.

If a sentence is classified by the annotator as non-salient, give it the label `n/a` (so that the model later learns to identify salient sentences within their context window).

STEP 2: MULTI-DIMENSIONAL VECTOR CLASSIFICATION. If a sentence is salient, do not pick a single category. Instead, evaluate the sentence against all four dimensions of psychological distance simultaneously to create a distance vector [Temporal, Spatial, Social, Hypothetic

DIMENSION AND DISTANCE MATRIX (OPERATIONAL BOUNDARIES)

For every salient sentence, assign a value for each dimension:

- **Value 0 (Near):** concrete, low-level cues present.
- **Value 1 (Far):** abstract, high-level cues present.
- **Value -1 (N/A):** no cues or information for this specific dimension.

Dimension	0: Near (low-level / concrete)	1: Far (high-level / abstract)
Temporal (When)	Current cycle: events within 1 month (e.g., "today", "last night").	Generation / era: events >1 month or indefinite (e.g., "in the 90s", "always").
Spatial (Where)	Travelable: local city, state, or buildings (e.g., "my house", "this office").	Map: national, global, or universal scale (e.g., "overseas", "the EU", "global").
Social (Who)	Individual: specific, identifiable singular people (e.g., "I", "the OP", "Elon Musk").	Categorical: groups, demographics, or institutions (e.g., "the police", "society", "the government").
Hypothetical	Facticity: objective reality, physical laws, history (e.g., "it is a fact", "data shows").	Thought experiment: possibilities or "what-ifs" (e.g., "suppose", "imagine", "ideally").

For Hypothetical, we annotate with -1 when a comment mentions a direct personal experience or observation (e.g., "I saw X", "I did Y", "I personally experienced Z"). These are not hypothetical.

THE "SPLIT" RULES (RESOLVING BORDERLINE CASES)

To ensure high inter-annotator agreement, two tests are applied.

- **The "nouns vs. adjectives" test.** Near distance relies on proper nouns (names, specific dates). Far distance relies on adjectives and adverbs (mostly, global, idealistic).
- **The "why vs. how" test.** Near distance focuses on mechanics (*how* it happened). Far distance focuses on purpose (*why* it matters).

HANDLING NON-DIMENSIONAL ARGUMENTS

Each dimension is coded independently as $\{-1, 0, 1\}$; there is no combined "mixed" value. A sentence that carries cues at different levels on different dimensions simply receives the corresponding value on each (for example, a near social reference inside a far hypothetical clause is coded social $= 0$, hypothetical $= 1$). The N/A value (-1) is used per dimension for any dimension on which a salient sentence carries no distance cue—for instance a pure logical connector ("therefore, the logic follows that...") is N/A on all four.

SALIENT SENTENCE EXAMPLES (VECTORISED)

WORKFLOW IN LABEL STUDIO

1. **Contextual reading.** Read the entire post first to understand the overarching argument.
2. **Salience filtering.** Identify sentences that function as a claim, evidence, or reasoning. For any sentence that is meta-talk or context, assign the label `n/a`.
3. **Vector verification.** For salient sentences, review the language-model-generated vector $[T, S, \mathrm{Soc}, H]$.
4. **Boundary audit.** Check the specific values against the operational boundaries (e.g., verify the 1-month rule for any T:0 or T:1 values).

Sentence	T	S	Soc	H
"When we exist 24/7 with these machines and internet, we lose regular humanity."	Near	—	Far	Near
"Lightning will strike me if I stay out in the rain."	—	Near	—	Far
"Will AI govern the world?"	—	Far	Far	Far
"It also might integrate with us very subtly over time, for example most people have mobile phones, wearables are coming out [. . .]"	Far	—	Far	Far
"I'm not worried about robots attacking humans, just like Vancouver Island Marmots don't generally need to be worried about humans [. . .]"	—	Near	Far	Near

Table 8: Example salient sentences with their coded distance vectors (T, S, Soc, H). "—" denotes N/A on that dimension. Drawn from the annotated set; the full few-shot set used by the assist prompt is in Appendix B.

5. **Finalise.** Confirm the final vector. If a dimension has no relevant cues, ensure it is set to -1.

B GPT-5-NANO ANNOTATION-ASSIST PROMPTS

GPT-5-nano was used to produce initial label predictions that annotators independently reviewed and corrected. Two prompt versions were used: an initial version for the first ~80 annotations (Section B.1) and a revised version applied from that point onward (Section B.2). The revision followed the Round 1 inter-annotator analysis, which showed the most systematic disagreement on the Temporal and Hypothetical dimensions; the operational definitions of these two dimensions were tightened in both the annotator guidelines (Appendix A) and the revised prompt below. The Round 2 inter-annotator agreement reported in Section 3.2.3 reflects annotation produced under the revised prompt.

B.1 ROUND 1 PROMPT (ANNOTATIONS 1–80)

```
SYSTEM_PROMPT = """You are a research assistant specializing in
Psycholinguistics and Construal Level Theory (CLT).

Your task is to analyze sentences from ChangeMyView (CMV) Reddit posts
with Psychological Distance vectors. You will receive the full post or
comment for context, and a numbered list of sentences to analyze.

## Your analysis task

For each sentence, perform two steps:

### Step 1 -- Salience filter
Classify the sentence as one of:
- "salient": contains a Claim, Evidence, or Reasoning
- "n/a": meta-talk, filler, greetings, transitions with no
  argumentative content

### Step 2 -- Distance vector (salient sentences only)
Assign a suggested vector [Temporal, Spatial, Social, Hypothetical].
Each dimension takes exactly one value: 0, 1, or -1.

Scoring:
- 0 (Near): concrete, specific, low-level construal
- 1 (Far):  abstract, categorical, high-level construal
- -1 (N/A): no cue present for this dimension

## Operational boundaries (apply strictly)

| Dimension    | 0 = Near                              | 1 = Far
                            |
|--------------|---------------------------------------|-----------------------------------------------|
```

```
| Temporal     | Events within 1 month                  | Events >1 month or indefinite               |
|              | ("today", "last week")                 | ("always", "in the 90s")                   |
| Spatial      | Local/travelable                       | National/global/universal                   |
|              | ("my house", "this city")              | ("the EU", "overseas")                     |
| Social       | "I", specific named individuals        | Groups, institutions                       |
|              |                                        | ("the government", "society")              |
| Hypothetical | Objective facts, history,              | Possibilities, what-ifs                    |
|              | physical laws                          | ("suppose", "ideally", "if X would Y")       |

## Output format
Return ONLY a valid JSON array. No prose, no markdown fences. One
object per sentence:

For a NON-SALIENT sentence:
{
 "sentence_idx": 0,
 "salient": false,
 "confidence": 3,
 "reasoning": "Brief explanation of why this sentence is not salient.",
 "uncertainty_note": ""
}

For a SALIENT sentence:
{
 "sentence_idx": 1,
 "salient": true,
 "vector": {"temporal": 1, "spatial": -1, "social": 1, "hypothetical": 0},
 "confidence": 2,
 "flag": "ok",
 "reasoning": "Brief explanation of the sentence content and why it is salient.",
 "vector_explanation": {
   "temporal": "Far (1) -- uses abstract policy language with no specific timeframe.",
   "spatial": "N/A (-1) -- no spatial cue present.",
   "social": "Far (1) -- refers to a general group rather than specific individuals.",
   "hypothetical": "Near (0) -- stated as a general fact, not a hypothetical."
 },
 "uncertainty_note": "Temporal is ambiguous -- could be near if interpreted as current situation."
}

Confidence: 3=certain, 2=one dimension ambiguous, 1=uncertain (flag="review").
Return the full array. Do not truncate."""
```

B.2 ROUND 2 PROMPT (REVISED, USED FROM ANNOTATION 81 ONWARD)

```
You are a research assistant specializing in Psycholinguistics and
Construal Level Theory (CLT).
Your task is to analyze sentences from ChangeMyView (CMV) Reddit posts
with Psychological Distance vectors. You will receive the full post or
comment for context, and a numbered list of sentences to analyze.

## Your analysis task
For each sentence, perform two steps:

### Step 1 -- Salience filter
Classify the sentence as one of:
- "salient": contains a Claim, Evidence, or Reasoning
- "n/a": meta-talk, filler, greetings, transitions with no
  argumentative content

### Step 2 -- Distance vector (salient sentences only)
Assign a suggested vector [Temporal, Spatial, Social, Hypothetical].
Each dimension takes exactly one value: 0, 1, or -1.

Scoring:
- 0 (Near): concrete, specific, low-level construal
- 1 (Far):  abstract, categorical, high-level construal
- -1 (N/A): no cue present for this dimension

## Operational boundaries (apply strictly)

| Dimension    | -1 = N/A (DEFAULT if no explicit cue)              | 0 = Near
                                                                  | 1 = Far
                                                                      |
|--------------|----------------------------------------------------|-----------------------------------------

| Temporal     | No time reference; laws, norms, or logic           | Explicit
     time anchor <= 1 month                                       | Explicit
    time anchor > 1 month or open-ended                                |
|              | with no stated timeframe                           | ("today",
     "last week", "right now", "currently happening")           | ("in the
    1990s", "for centuries", "always", "over the next decade")         |
| Hypothetical | Statements of direct personal experience           | Asserted
     facts, historical events, established findings stated as     |
    Possibilities, conditionals, what-ifs, ideals, or speculation
           |
|              | or first-hand observation ("I did X", "I saw Y")   | true ("
    studies show", "X happened in 1990")                           | ("if X
    then Y", "suppose", "ideally", "could", "imagine if")
    |

### Critical rules for Temporal and Hypothetical

**Temporal**: Laws, policies, institutions, and general norms have NO
inherent temporal distance -- assign -1 unless the sentence contains
```

```
an explicit time expression. "The law requires X" -> -1. "The law
passed last month" -> 0. "This has been the law for decades" -> 1.
The question is always: does this sentence contain a time word or
phrase? If not, -1.

**Hypothetical**: This dimension captures whether the content is
presented as real/actual vs. possible/imagined. It is NOT about
whether the sentence sounds abstract or philosophical.
"Corporations lobby for profit" -> 0 (asserted as fact).
"If corporations were regulated, they would lobby less" -> 1.
"I personally witnessed this" -> -1.
When in doubt, ask: is the author asserting this as true, or imagining
a scenario?

| Dimension | -1 = N/A                  | 0 = Near                                          | 1 = Far                             |
|-----------|---------------------------|---------------------------------------------------|-----------
| Spatial   | No location reference     | Local/travelable                                  | National/global/universal           |
|           |                           | ("my house", "this city", "our neighborhood")     | ("the EU", "overseas", "across the country") |
| Social    | No social reference       | "I", specific named individuals                   | Groups, institutions                |
|           |                           |                                                   | ("the government", "society", "women", "immigrants") |

## Output format
Return ONLY a valid JSON array. No prose, no markdown fences. One
object per sentence:

For a NON-SALIENT sentence:
{
 "sentence_idx": 0,
 "salient": false,
 "confidence": 3,
 "reasoning": "Brief explanation of why this sentence is not salient.",
 "uncertainty_note": ""
}

For a SALIENT sentence:
{
 "sentence_idx": 1,
 "salient": true,
 "vector": {"temporal": 1, "spatial": -1, "social": 1, "hypothetical": 0},
 "confidence": 2,
 "flag": "ok",
 "reasoning": "Brief explanation of the sentence content and why it is salient.",
 "vector_explanation": {
   "temporal": "Far (1) -- uses abstract policy language with no specific timeframe.",
   "spatial": "N/A (-1) -- no spatial cue present.",
   "social": "Far (1) -- refers to a general group rather than specific individuals.",
   "hypothetical": "Near (0) -- stated as a general fact, not a hypothetical."
```

```
 },
 "uncertainty_note": "Temporal is ambiguous -- could be near if interpreted as
     current situation."
}

Confidence: 3=certain, 2=one dimension ambiguous, 1=uncertain (flag="review").
Return the full array. Do not truncate.
```

C LLAMA-3.2-3B SHOOTOUT PROMPTS

The prompted LLaMA-3.2-3B-Instruct cascade was the third candidate contributed by this thesis to the classifier shootout (Section 5.1). Inference used greedy decoding (`do_sample=False`) for reproducibility. Few-shot examples were drawn from the training split only; their task IDs are hard-coded in the configurations below so that examples do not drift between runs. The Stage 2 examples were hand-selected to ensure that every one of the twelve (dimension $\times$ class) combinations appears at least once, with deliberate over-representation of the rare Near/Far classes on Temporal and Spatial relative to the training distribution.

C.1 STAGE 1 (SALIENCE) CONFIGURATION

```yaml
# LLaMA-3.2-3B-Instruct Stage 1 (salience) config
#
# parse_failure_fallback: 0 (non-salient).
# Salient is actually the MAJORITY class in train (621/723 = 85.9%),
# so this fallback intentionally chooses the MINORITY class. The
# rationale is downstream-cost asymmetry: a parse failure on Stage 1
# should not push a sentence into the expensive Stage 2 path
# spuriously. We accept a small recall loss on Stage 1 to avoid
# poisoning the Stage 2 evaluation set with sentences whose salience
# is unknown.

model_name: meta-llama/Llama-3.2-3B-Instruct

prompt_template_version: "v1.1"

# Few-shot example task_ids drawn from train, hard-coded for reproducibility.
few_shot_task_ids:
  - {task_id: 13264, sentence_idx: 5,  label: 0}   # filler transition
  - {task_id: 13264, sentence_idx: 0,  label: 1}   # claim
  - {task_id: 13282, sentence_idx: 29, label: 1}   # concrete reasoning
  - {task_id: 13267, sentence_idx: 9,  label: 0}   # non-salient transition
  - {task_id: 13270, sentence_idx: 1,  label: 1}   # claim with abstract group cues

# System prompt injected as the system role in the chat template.
system_prompt: |
  You are a research assistant specializing in argument analysis. Your
  task is to classify whether a sentence from an online argument is
  "salient" -- that is, whether it contains the main claim, supporting
  evidence, or connecting reasoning of the argument.

  Non-salient sentences are filler, transitional phrases,
  meta-commentary, or conversational acknowledgements that carry no
```

```
  argumentative content.

  Reply with exactly one token: 1 if the sentence is salient, 0 if it
  is not. Do not add any explanation, punctuation, or other text.

# User-turn template. {few_shot_block} and {sentence} are filled at runtime.
user_prompt_template: |
  {few_shot_block}Sentence: "{sentence}"
  Salient (1) or non-salient (0)?

# Template for each few-shot example inside the user turn.
few_shot_example_template: |
  Sentence: "{sentence}"
  Salient (1) or non-salient (0)? {label}

# Generation: greedy decoding hard-coded in llama_inference.py.
max_new_tokens: 1

# Parse-failure fallback: 0 = non-salient (minority class).
parse_failure_fallback: 0
```

C.2 STAGE 2 (CLT VECTOR) CONFIGURATION

```
# LLaMA-3.2-3B-Instruct Stage 2 (CLT dimensions) config
#
# parse_failure_fallback: -1 on every dimension (N/A) -- majority class.
#   Temporal N/A: 90.5%  Spatial N/A: 92.6%  Social N/A: 18.0%  Hypothetical N
    /A: 6.6%
# Falling back to all-N/A is conservative and avoids hallucinating rare
# Near/Far signals when the model output is malformed.
#
# Few-shot selection rationale: the training distribution is severely
# skewed (Temporal Near 3.7%, Temporal Far 5.8%, Spatial Near 4.3%,
# Spatial Far 3.1%). Without deliberate over-representation the model
# would see mostly N/A and infer N/A is always correct. The 8 examples
# below are hand-picked to ensure every one of the 12 dimension x class
# combinations appears at least once.

model_name: meta-llama/Llama-3.2-3B-Instruct

prompt_template_version: "v1.0"

# 8 few-shot examples, all from train; verified against train.json
# on 2026-05-09.
few_shot_task_ids:
  # Example 1 -- temporal Far, spatial N/A, social Far, hypothetical Near
  - task_id: 13264
    sentence_idx: 0
    temporal: 1
    spatial: -1
```

```
    social: 1
    hypothetical: 0

  # Example 2 -- temporal N/A, spatial Far, social Far, hypothetical Near
  - task_id: 13264
    sentence_idx: 13
    temporal: -1
    spatial: 1
    social: 1
    hypothetical: 0

  # Example 3 -- temporal Near, spatial N/A, social Far, hypothetical Near
  - task_id: 13270
    sentence_idx: 1
    temporal: 0
    spatial: -1
    social: 1
    hypothetical: 0

  # Example 4 -- temporal N/A, spatial Near, social Far, hypothetical Near
  - task_id: 13266
    sentence_idx: 2
    temporal: -1
    spatial: 0
    social: 1
    hypothetical: 0

  # Example 5 -- temporal Far, spatial N/A, social Far, hypothetical Far
  - task_id: 13270
    sentence_idx: 0
    temporal: 1
    spatial: -1
    social: 1
    hypothetical: 1

  # Example 6 -- temporal N/A, spatial Near, social N/A, hypothetical Far
  - task_id: 13275
    sentence_idx: 4
    temporal: -1
    spatial: 0
    social: -1
    hypothetical: 1

  # Example 7 -- temporal Near, spatial N/A, social Near, hypothetical N/A
  - task_id: 11353
    sentence_idx: 0
    temporal: 0
    spatial: -1
    social: 0
    hypothetical: -1

  # Example 8 -- temporal N/A, spatial Far, social Far, hypothetical Far
  - task_id: 13267
    sentence_idx: 14
    temporal: -1
    spatial: 1
    social: 1
    hypothetical: 1

system_prompt: |
  You are a research assistant specializing in Construal Level Theory
```

```
  (CLT) and psycholinguistics. You will receive a salient sentence from
  an online argument and must classify its psychological distance
  across four dimensions simultaneously.

  For each dimension output one of three integer values:
    -1  =  N/A   -- no cue for this dimension is present in the sentence
     0  =  Near  -- concrete, low-level, specific cue present
     1  =  Far   -- abstract, high-level, categorical cue present

  Dimension definitions:
    temporal    : WHEN  -- Near if < 1 month or "now/today";
                           Far if > 1 month or era-level
    spatial     : WHERE -- Near if local/travelable (city, building, "here");
                           Far if national/global
    social      : WHO   -- Near if a named individual or "I/my";
                           Far if a group/institution/society
    hypothetical: HOW REAL -- Near if factual/historical;
                               Far if "what-if"/possibility/supposition

  Output ONLY a single JSON object on one line, no prose, no explanation:
  {"temporal": <int>, "spatial": <int>, "social": <int>, "hypothetical": <int>}

user_prompt_template: |
  {few_shot_block}Sentence: "{sentence}"
  CLT vector:

few_shot_example_template: |
  Sentence: "{sentence}"
  CLT vector: {{"temporal": {temporal}, "spatial": {spatial}, "social": {social}, "hypothetical": {hypothetical}}}

# Generation: greedy decoding hard-coded in llama_inference.py.
max_new_tokens: 80

# Parse-failure fallback: all dimensions to N/A (majority class).
parse_failure_fallback:
  temporal: -1
  spatial: -1
  social: -1
  hypothetical: -1
```

C.3 EXPLORATORY DESCRIPTIVES: TEMPORAL AND SPATIAL INDICES

Table 9: Index availability and distribution on the full inference corpus (n=598,272). Means are computed over defined (non-NA) values only. The Δ-group split is reported for Temporal and Spatial only; Social and Hypothetical enter the confirmatory regression and are not split here. No inferential test is performed.

Index	% def.	Mean	Median	SD	Mean (Δ=1)	Mean (Δ=0)	n defined
Temporal	38.2	0.653	1.00	0.424	0.665	0.653	228,371
Spatial	25.7	0.503	0.50	0.472	0.511	0.502	153,735
Social	93.2	0.671	0.80	0.368	—		557,529
Hypothetical	95.4	0.395	0.33	0.348	—		570,816

C.4 SURFACE CONTROL COEFFICIENTS (MODEL 1)

Table 10 reports the coefficients on the nine surface controls in the adjusted primary model (M1, Section 5.5). All controls are $\ln(x+1)$-transformed. Estimates are log-odds with thread-clustered robust standard errors. These are adjustment covariates and are not interpreted as effects of theoretical interest; they are reported for transparency.

Table 10: Coefficients on the nine $\ln(x+1)$-transformed surface controls in the adjusted conditional logistic model (M1), primary 1:3 sample. Log-odds with thread-clustered robust SEs.

Control (log-transformed)	$\hat{\beta}$	Robust SE	z	p
Noun-to-verb ratio	0.225	0.078	2.90	.004
Type–token ratio	−8.427	0.342	−24.61	$< .001$
Flesch–Kincaid grade	−0.498	0.139	−3.59	$< .001$
Hedge density	−1.429	0.990	−1.44	.149
Mean sentence length	0.436	0.106	4.10	$< .001$
Mean word length	5.756	0.424	13.58	$< .001$
Punctuation density	−6.793	2.353	−2.89	.004
Paragraph count	0.320	0.046	6.95	$< .001$
Number of URLs	0.511	0.053	9.58	$< .001$

C.5 DESCRIPTIVE STATISTICS FOR THE SURFACE CONTROLS

Table 11 reports descriptive statistics for the nine surface controls on the matched analysis sample ($n = 23{,}096$), before the $\ln(x+1)$ transform applied in estimation.

Table 11: Raw (pre-transform) descriptives for the nine surface controls on the matched 1:3 analysis sample ($n = 23{,}096$).

Control	Mean	SD	Median	Min	Max
Noun-to-verb ratio	1.640	0.871	1.500	0.000	24.000
Type–token ratio	0.696	0.132	0.696	0.259	1.000
Flesch–Kincaid grade	9.693	3.357	9.351	−0.006	111.591
Hedge density	0.024	0.020	0.021	0.000	0.205
Mean sentence length	19.372	7.715	18.148	3.500	288.000
Mean word length	4.429	0.374	4.417	2.655	8.188
Punctuation density	0.027	0.009	0.026	0.000	0.173
Paragraph count	3.390	3.335	2.000	1.000	65.000
Number of URLs	0.197	0.789	0.000	0.000	31.000

10.9% of comments contain at least one URL.

C.6 RUNNER-UP CASCADE: ROBERTA PER-DIMENSION DIAGNOSTICS

Table 12 reports the per-dimension test-set performance of the RoBERTa-base cascade, the runner-up in the classifier shootout (Section 5.1). Unlike the selected BERT-base cascade, these figures are from the development-trained shootout checkpoint rather than a model retrained on the train+dev union, and are included to document where the two encoders diverge. RoBERTa recovers the Spatial dimension noticeably better than BERT ($\alpha = 0.51$ vs. 0.20), while both collapse on Temporal.

Table 12: Per-dimension performance of the RoBERTa-base cascade on the held-out test set ($n = 127$ salient sentences), shootout (development-trained) checkpoint. Per-class F_1 is given for N/A, Near, and Far. Stage 1 (salience): macro-$F_1 = 0.842$, accuracy 0.897, $\alpha = 0.686$.

Dimension	Macro-F_1	Kripp. α	Per-class F_1 (N/A, Near, Far)	Status
Temporal	0.309	0.031	(0.83, 0.00, 0.10)	Collapsed
Spatial	0.560	0.509	(0.91, 0.18, 0.59)	Weak (Near sparse)
Social	0.472	0.221	(0.11, 0.49, 0.82)	Usable
Hypothetical	0.536	0.264	(0.45, 0.81, 0.35)	Usable

C.7 MULTICOLLINEARITY SCREEN (VIF)

Before estimation, the candidate control set was screened for multicollinearity using variance inflation factors (VIF). Table 13 reports the screen. Three length-related variables (raw token count, word-token count, and sentence count) exceeded the conventional VIF > 10 threshold and were dropped; mean sentence length, although marginal in the initial screen, falls below the threshold once the three collinear counts are removed and is retained. The two abstraction indices of theoretical interest (A_{Soc}, A_H) show no collinearity concern.

Table 13: Variance inflation factors for the candidate predictor set. "Drop" marks variables removed for VIF > 10; the nine retained controls enter Model 1.

Variable	VIF	Decision
Word-token count	1564.86	Drop
Token count	1527.62	Drop
Sentence count	62.41	Drop
Mean sentence length	12.33	Retained†
Type–token ratio	6.09	Retained
Flesch–Kincaid grade	5.53	Retained
Punctuation density	3.29	Retained
Mean word length	2.94	Retained
Paragraph count	2.79	Retained
Social abstraction index (A_{Soc})	1.33	Retained
Hypothetical abstraction index (A_H)	1.27	Retained
Noun-to-verb ratio	1.19	Retained
Number of URLs	1.11	Retained
Hedge density	1.09	Retained

†VIF falls below 10 once the three collinear count variables are dropped.

C.8 CONVERGENCE OF THE ABSTRACTION INDICES

Table 14 reports the pairwise Pearson correlations among the four comment-level abstraction indices, computed over the full inference corpus with pairwise-complete observations (each pair uses all comments for which both indices are defined, so n varies by pair). The two indices used in the confirmatory analysis, A_{Soc} and A_H, are weakly negatively correlated; the remaining pairs involve the sparse Temporal and Spatial indices. A principal-component analysis of A_{Soc} and A_H (standardised) returns a first component that is a contrast rather than a shared factor (loadings $A_{\text{Soc}} = +0.71$, $A_H = -0.71$; 64.2% of variance), consistent with the negative correlation.

Table 14: Pairwise Pearson correlations among the four comment-level abstraction indices on the full inference corpus, pairwise-complete. All $p < .001$. n is the number of comments with both indices defined.

Index pair	r	n
$A_{\text{Soc}} - A_H$	-0.28	543,740
$A_T - A_{\text{Soc}}$	$+0.07$	227,400
$A_T - A_H$	-0.15	218,573
$A_T - A_S$	$+0.04$	128,141
$A_S - A_{\text{Soc}}$	$+0.49$	153,292
$A_S - A_H$	-0.20	147,045

C.9 CONSTRUAL FIT: FULL MODEL TABLE

Table 15 reports the three fit specifications of Section 4.3.3 at the primary 1:3 and robustness 1:5 matching ratios. Only the construal and fit terms are shown; the nine surface controls are included in every model and behave as in Table 10. Odds ratios are over the $[0, 1]$ index range, with two-tailed p from thread-clustered robust standard errors. The OP main-effects model (F1) is reported for completeness: A^{OP}_{Soc} and A^{OP}_H are constant within a thread and absorbed by the conditional likelihood, so they are not estimable and F1 reduces to the comment-only specification.

Table 15: Fit specifications F1–F3 (Section 4.3.3) at the 1:3 and 1:5 matched samples. Odds ratios with two-tailed p from thread-clustered robust standard errors. OP main effects are not identified (see text). $n = 23{,}068$ (1:3) and 33,850 (1:5); 5,842 strata in both.

	1:3		1:5	
Term	OR	p	OR	p
F1: OP main effects				
A_{Soc}	0.857	.015	0.861	.012
A_H	0.898	.105	0.863	.016
$A^{OP}_{\text{Soc}}, A^{OP}_H$	not identified			
F2: interaction				
A_{Soc}	0.561	$< .001$	0.597	$< .001$
A_H	0.825	.100	0.815	.052
$A_{\text{Soc}} \times A^{OP}_{\text{Soc}}$	1.914	.002	1.756	.005
$A_H \times A^{OP}_H$	1.251	.427	1.164	.551
F3: distance (mismatch)				
A_{Soc}	0.769	$< .001$	0.788	$< .001$
A_H	0.941	.401	0.907	.143
$\lvert A_{\text{Soc}} - A^{OP}_{\text{Soc}} \rvert$	0.642	$< .001$	0.685	$< .001$
$\lvert A_H - A^{OP}_H \rvert$	0.794	.015	0.793	.008